

Overclocking the p-bit: Stochastic Binary Neural Network Inference with Autocorrelated sMTJ Randomness

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Abstract—Stochastic magnetic tunnel junctions (sMTJs) supply physically random bits at ~ 0.03 pJ per bit, five to six orders of magnitude below software pseudorandom generation, making them natural entropy sources for probabilistic-bit (p-bit) neural hardware. A standard design rule requires sampling an sMTJ no faster than twice its mean dwell time so that successive bits are uncorrelated; existing systems obey it by construction, and its cost to an application has not been measured. Treating the sampling-to-correlation-time ratio as a continuous design knob, we simulate stochastic binary network classification on MNIST and Fashion-MNIST with telegraph-process sMTJ neurons and measure where accuracy actually fails. Operating mode, not sampling rate, is decisive. With a per-input settle of ≈ 4 correlation times—a requirement we measure, not assume—accuracy is nearly rate-independent: with 32-sample averaging, $200\times$ faster-than-rule sampling costs under 0.7 accuracy points. In continuous streaming operation, stale device state carries each input’s activations into the next, and single-sample accuracy collapses to chance. Device-to-device dwell-time dispersion does not rescue streaming: at a fixed central fluctuation rate it is mildly harmful, and the apparently decisive choice is which population statistic fabrication pins—fixing the mean rather than the median correlation time silently multiplies every device’s rate by $e^{\sigma_\Delta^2/2}$, a $3\times$ effect at $\sigma_\Delta = 1.5 k_B T$ that outweighs dispersion itself. Finally, correlation-aware training—backpropagating while activations are sampled from the correlated process—recovers most of the streaming loss (28% $\rightarrow$ 85% at the harshest operating point, for a 3.3-point cost at nominal speed), extending the accuracy-throughput Pareto frontier to $2\times$ the rule-compliant optimum at 97% accuracy, $4\times$ at 95%, and $8\times$ at 90%.

Index Terms—probabilistic computing, p-bit, stochastic magnetic tunnel junction, binarized neural networks, random telegraph noise, energy-efficient inference

I. INTRODUCTION

GENERATING randomness is expensive on conventional hardware: pseudorandom generation at Gb/s rates costs ~ 30 nJ per bit on CPUs and ~ 3 nJ per bit on GPUs [1]. A stochastic magnetic tunnel junction (sMTJ)—a nanomagnet whose free layer flips between two states under thermal agitation—delivers physically random bits at ~ 0.03 pJ per bit [1]. This $\sim 10^6$ gap motivates probabilistic-bit (p-bit) computing [2], [3], in which the sMTJ is not a peripheral generator but the stochastic neuron itself: biased by an input current, its time-averaged state follows $\langle m \rangle = \tanh(I)$ [2]—the activation of stochastic binary neural networks (SBNNs).

Every physical entropy source has a clock. An sMTJ flips on its own schedule; read faster than it flips, it returns copies

of a stale state rather than fresh samples. Daniels *et al.* showed that the autocorrelation between adjacent samples vanishes once the sampling interval exceeds about twice the *mean dwell time* τ_d [4], and this “ 2τ rule” functions as a hard constraint: system demonstrations oversample device relaxation times by large factors [5], and randomness quality is certified against statistical test suites [1], [6]. The clock-versus-device-timescale trade has been characterized for sampling machines: autonomous Ising-machine operation requires synapse response faster than neuron fluctuation [7], and in situ Boltzmann learning requires the learning time constant to exceed the MTJ fluctuation time [8]. What is missing is the *application-side price curve*: no published work maps the sampling ratio continuously onto feedforward classification accuracy, distinguishes the operating modes in which that price is or is not paid, or asks what training can buy back. This paper does. Concretely, our contributions are:

- **Mode, not rate, is decisive.** We simulate sMTJ neurons as input-biased telegraph processes (exact discrete-time propagator; stationary law equal to the p-bit activation) and sweep the sampling ratio $r_0 = \Delta t / \tau_{\text{corr}}(0)$ across four decades. With a per-input settle of $\approx 4\tau_{\text{corr}}$ —a requirement we *measure* by sweeping the settle interval, replacing the folklore idle assumption—classification is nearly rate-independent: at $r_0 = 0.02$, i.e. $200\times$ past the rule, 32-sample accuracy drops only 97.7% $\rightarrow$ 97.1%. In continuous streaming operation, stale state leaks each input’s activations into the next input’s samples and single-sample accuracy collapses to chance (10%) below $r_0 \approx 0.05$; sample-averaging buys roughly an order of magnitude in r_0 per $32\times$ increase in samples. The collapse persists (indeed worsens) under a bias-independent-rate device variant, so it is generic staleness, not an artifact of the Arrhenius parameterization.
- **The centering convention, not dispersion, governs variability.** Whether device-to-device dwell-time dispersion appears to hurt or help streaming is decided by which population statistic fabrication holds fixed. Because dwell time is exponential in the barrier offset, fixing the *mean* correlation time rather than the *median* device multiplies every device’s rate by $e^{\sigma_\Delta^2/2}$ ($3.1\times$ at $\sigma_\Delta = 1.5 k_B T$); this uniform offset explains essentially all of the apparent rescue (72% $\rightarrow$ 94% at $r_0 = 0.05$) through the baseline accuracy-vs- r_0 curve, while dispersion at a matched central rate costs 1–5 points (the slow tail dominates).

A device specification must therefore pin the central fluctuation rate; dispersion around it is a second-order penalty.

- **Correlation-aware training extends the Pareto frontier.** Training while activations are sampled from the same correlated telegraph process (state running continuously across minibatches) recovers most of the streaming loss: 28% $\rightarrow$ 85% at $r_0 = 0.02$ ($T=32$), at a 3.3-point cost at nominal speed (a milder variant costs 1.2 points). With physically-accounted throughput (settle mode pays its measured $\approx 4\tau_{\text{corr}}$ per input), the accuracy–throughput frontier reaches $2\times$ the best rule-compliant operating point at 97% accuracy, $4\times$ at 95%, and $8\times$ at 90%—the latter two from training alone.

II. BACKGROUND AND RELATED WORK

A. *p*-bits and sMTJ entropy sources

A p-bit outputs $m = \text{sgn}\{\text{rand}(-1, 1) + \tanh I\}$ per query, so $\langle m \rangle = \tanh(I)$ [2]; superparamagnetic MTJs with barriers near $1.4 k_B T$ [1] realize this natively, with dwell times $\tau = \tau_0 e^{\Delta/k_B T}$, $\tau_0 \approx 10 \text{ ps} - 1 \text{ ns}$ [2]. Demonstrated fluctuation timescales span nanoseconds—2–5 ns autocorrelation in easy-plane devices [9], ns-scale telegraph noise in in-plane junctions [10]—to the microsecond-and-slower devices of early p-bit demonstrations [11], whose system used $\sim \text{ms}$ -class fluctuations; a recent heterogeneous CMOS-plus-nanomagnet system oversamples its $\sim 20 \text{ ms}$ relaxation by $40\times$ [5].

B. Randomness quality, sampling, and neural inference

Robustness studies of MTJ neural hardware concentrate on *static* imperfections: process-variation-induced sigmoid distortion [12], [13] and fabricated-array parameter spread with compensation [14]. sMTJ generators are validated with NIST statistical suites [1], [6]; software-PRNG quality shows a threshold effect on MNIST training [15]; measured sMTJ bitstreams have driven stochastic-computing inference with whatever correlation the apparatus had [16]. The stochastic-computing literature has long recognized bit-level correlation as a first-order design variable [17], [18], and multi-sample averaging of stochastic binary activations—including training through the average—was recently shown to improve p-bit network accuracy [19]. On the theory side, the cost of correlated samples is classical: autocorrelation time divides the effective sample size of any Monte Carlo average [20], and noise level has a computational optimum in spiking networks [21]. Our contribution sits at the unoccupied intersection: the sampling ratio as a continuously swept variable, mapped to feedforward classification accuracy, across operating modes, device-variability models, and training regimes.

III. MODEL

A. The sMTJ neuron as a biased telegraph process

Each hidden unit is a two-state fluctuator $s \in \{-1, +1\}$ with input-modulated Arrhenius escape rates

$$k_{-\rightarrow+} = f e^{+I}, \quad k_{+\rightarrow-} = f e^{-I}, \quad (1)$$

where I is the dimensionless synaptic input and $f = f_0 e^{-\delta\Delta}$ the device attempt rate with per-device barrier offset $\delta\Delta$ (in $k_B T$). The stationary law is

$$P(s=+1) = \frac{k_{-\rightarrow+}}{k_{-\rightarrow+} + k_{+\rightarrow-}} = \frac{1 + \tanh I}{2} \equiv p, \quad (2)$$

which follows from $\langle m \rangle = \tanh I$ [2] since $\langle m \rangle = 2P(+1) - 1$: the mean behavior of every neuron is exactly the p-bit activation at every operating point; only the time structure changes. The correlation time is

$$\tau_{\text{corr}}(I) = \frac{1}{k_{-\rightarrow+} + k_{+\rightarrow-}} = \frac{1}{2f \cosh I}, \quad (3)$$

so saturated neurons decorrelate faster. Note the distinction between τ_{corr} and the mean dwell time: at $I=0$, $\tau_d = 1/f = 2\tau_{\text{corr}}$, so the Daniels $2\tau_d$ rule [4] corresponds to $r_0 \geq 4$ in our units (at bias the occupancy-weighted dwell-to-correlation ratio grows as $2 \cosh 2I$, so in dwell-time units the rule tightens for driven devices; we quote it at $I=0$),

$$r_0 = 2f_0 \Delta t = \Delta t / \tau_{\text{corr}}(0) \big|_{\delta\Delta=0}. \quad (4)$$

Under the assumption that inputs are piecewise-constant over each inter-query interval (sample-and-hold operation between clocked layers), the exact two-state propagator applies with no discretization error:

$$P(s_{t+\Delta t}=+1 \mid s_t) = p + (\mathcal{K}[s_t=+1] - p) e^{-\Delta t / \tau_{\text{corr}}}, \quad (5)$$

with $\Delta t / \tau_{\text{corr}} = r_0 e^{-\delta\Delta} \cosh I$.

Eq. (1) is the energy-tilt picture appropriate for barrier-type (Arrhenius-regime) devices. Easy-plane sMTJs are engineered so the fluctuation rate is only weakly bias-dependent [1], [9]; we therefore also simulate a **bias-independent-rate variant** (total rate fixed at $2f$, giving $\Delta t / \tau_{\text{corr}} = r_0 e^{-\delta\Delta}$ with the same stationary law), and report where the two device classes differ. Device variability is modeled as $\delta\Delta \sim \mathcal{N}(\mu, \sigma_\Delta^2)$ fixed per chip. Because dwell time is exponential in $\delta\Delta$, the choice of μ decides which population statistic is held fixed as σ_Δ grows: $\mu = 0$ fixes the *median* device (the mean correlation time then grows as $e^{\sigma_\Delta^2/2}$), while $\mu = -\sigma_\Delta^2/2$ fixes the *mean* correlation time. We simulate both, since real fabrication corners could resemble either. In both cases every stationary firing probability is untouched, isolating the purely temporal effect of variability, complementary to static sigmoid-distortion studies [12]–[14]. Idealizations: no read-disturb, no $1/f$ resistance noise, no temperature drift, two-state telegraph regime [9], [10].

B. Operating modes: settle vs. streaming

In **settle mode** (batch-style inference), the input is applied and the device relaxes for an interval $t_s = \rho \tau_{\text{corr}}(0)$ before the first of T reads taken Δt apart. A long-idle device equilibrates to $p = 1/2$, *not* to the upcoming input's stationary law; in continuous operation the prior state is the previous input's final state. In either case the settle relaxes the device toward $p_{\text{eq}}(I)$ with time constant $\tau_{\text{corr}}(I)$: starting from idle, the first read carries a residual bias $-\frac{1}{2} \tanh(I) e^{-\rho \cosh I}$. We simulate the relaxation explicitly—the settle is a propagator step from

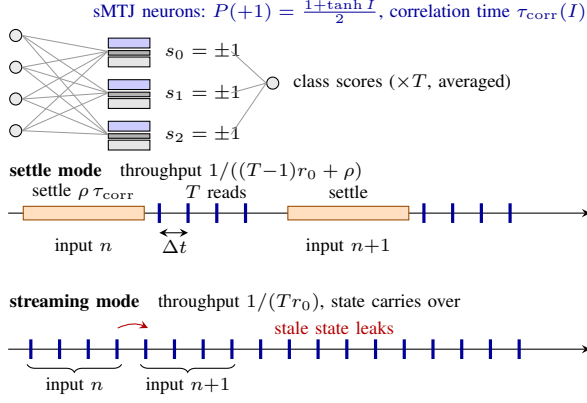


Fig. 1. SBNB with sMTJ neurons (top) and the two operating modes (bottom). Settle mode pays $\rho \tau_{\text{corr}}$ of relaxation per input; in streaming mode devices are never idle and the previous input’s activation pattern persists in the device states.

the device’s actual prior state, so what it erases is set by ρ , not assumed—and sweep ρ to find the requirement empirically. Throughput accounting charges ρ per input: settle-mode throughput is $1/((T-1)r_0 + \rho)$ images per τ_{corr} , bounded by $1/\rho$ however fast the reads are. In **streaming mode** the devices are never idle: state persists from each input to the next (throughput $1/(Tr_0)$). Note that a single settled read never invokes the inter-read interval at all: settle-mode $T=1$ accuracy depends only on ρ by construction, and its flatness across r_0 is a consistency check rather than a finding; the empirical content of settle mode is the ρ requirement and the cost of the $T>1$ correlated reads.

C. Network and training

We use a 784–256–128–10 MLP with stochastic bipolar hidden activations resampled at every query *including inference* (p-bit operating style, unlike test-time-deterministic binarized networks [22], [23]); real-valued input and readout layers follow standard practice. Class scores average T stochastic passes. Baselines train with ideal i.i.d. randomness and a straight-through estimator [24]; correlation-aware networks train identically but with hidden activations sampled from the telegraph model, device state running continuously across minibatches.

IV. EXPERIMENTAL SETUP

Datasets: MNIST [25] with Fashion-MNIST [26] replication. The last 5,000 training images form a validation split used for all model selection; the 10,000-image test set is used only for reported numbers. **Training:** 30 epochs, Adam (10^{-3} , cosine decay); ideal-randomness anchors reach 97.0% ($T=1$) and 97.8% ($T=32$) on MNIST. Sweeps cover $r_0 \in [0.02, 10]$ and $T \in \{1, \dots, 64\}$ with 3–5 thermal seeds per point (binomial SE on the 10k test set is ± 0.2 –0.5 points; error bars report seed std). Ideal-randomness anchors replicate across five independently trained networks, and key E2/E3 operating points were re-evaluated on four additional independently trained networks ([XNETS]); correlation-aware results average three

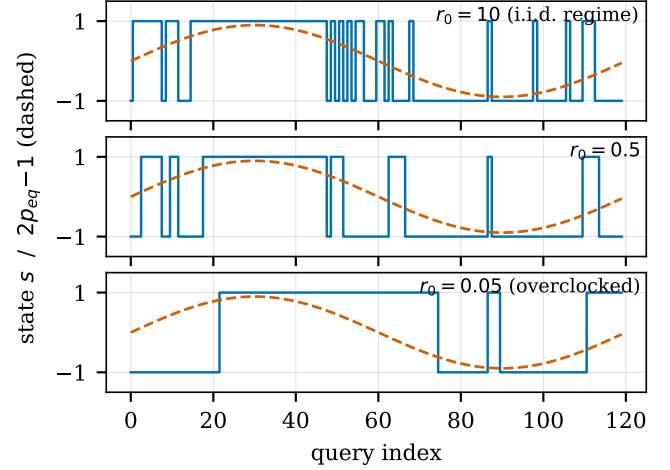


Fig. 2. Telegraph sMTJ neuron at three sampling ratios, driven by a slowly varying firing probability (dashed). At $r_0=10$ samples are effectively i.i.d.; at $r_0=0.05$ the device returns long runs of stale state.

training seeds per configuration (except $r_0^{\text{train}}=0.1$, one seed). Streaming is evaluated as true per-lane sequential streams: the test set is split into 100–200 independent lanes, each preceded by unscored warm-up images from the training pool so that scored images are in steady state; measured accuracy is invariant to lane count ([LANEINV] at 50/100/200/500 lanes for the same operating point and warm-up), confirming the protocol. Every result file records its git revision; runs are reused only when the simulator source is unchanged across the recorded revisions. The simulator’s stationary law, auto-correlation decay at zero and nonzero bias, settle-relaxation distribution, and variant behaviors are validated by unit tests against Eqs. (2)–(5). All experiments run on CPU in hours; code, per-run records, and configurations are in the project repository, public upon publication.

V. RESULTS

A. Mode, not rate (E2)

Fig. 3 is the central result. **Settle mode:** with $\rho = 4$, single-read accuracy is flat at 96.90% across the entire sweep—necessarily so, since a single settled read never invokes the inter-read interval (Sec. III-B); its exact flatness is a consistency check, not a finding. The empirical content is twofold. The settle sweep locates the actual re-equilibration requirement: single-read accuracy is 92.9%, 96.6%, 96.9%, 96.8% for $\rho = 1, 2, 4, 8$ —a settle of $\approx 4\tau_{\text{corr}}$ suffices and $\approx 1\tau_{\text{corr}}$ measurably does not. And the correlated-read cost is small: $T=32$ accuracy falls only 97.73% $\rightarrow$ 97.12% at $r_0=0.02$ —200 $\times$ past the rule. The residual $T=32$ loss is the effective-sample-size cost of correlated averaging [20]: stale within-input reads shrink T toward an effective T_{eff} , and the floor—the $T=1$ anchor—is high.

Streaming: steady-state accuracy collapses, and far harder than a warm-start-contaminated protocol would suggest. Single-shot accuracy is 96.7% at $r_0=2$, 92.7% at 1, 68.6% at 0.5, 27.5% at 0.2, 14.9% at 0.1, and chance ($\approx 10\%$) at

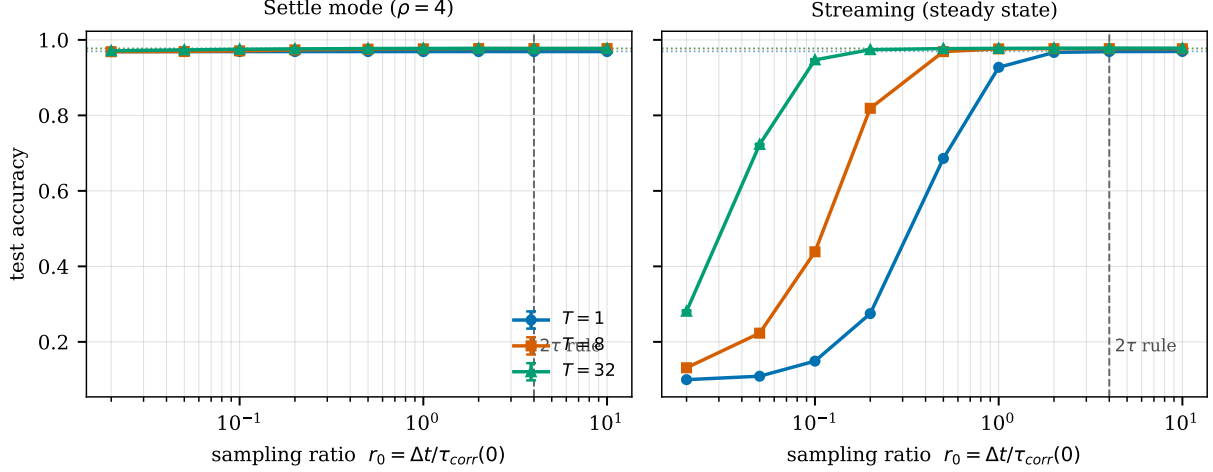


Fig. 3. Accuracy vs. sampling ratio r_0 (MNIST; mean $\pm$ std over thermal seeds; dotted: ideal-randomness anchors; dashed vertical: the $2\tau_d$ rule, $r_0 = 4$). **Left:** settle mode ($\rho=4$) is nearly rate-independent. **Right:** steady-state streaming collapses; $T=1$ reaches chance below $r_0 \approx 0.05$.

$r_0 \leq 0.05$: each input inherits the previous input’s activation pattern through the device state. Averaging buys roughly an order of magnitude in r_0 per $32\times$ increase in T (90%-crossings at $r_0 \approx 0.93, 0.33, 0.09$ for $T = 1, 8, 32$), but at $r_0 = 0.02$ even $T=32$ retains only 28.1%. The same qualitative structure holds for the bias-independent-rate variant—collapse thresholds shift roughly $2\times$ higher ($T=32$ falls to 53.8% already at $r_0=0.1$, vs. 94.7% for the Arrhenius device)—showing (i) the collapse is generic staleness, not a parameterization artifact, and (ii) the input-dependent rate boost $\cosh I$ of barrier-type devices is *protective*: strongly driven neurons refresh fastest exactly where it matters. Fashion-MNIST reproduces the full pattern: settle-mode $T=32$ falls only 88.2% $\rightarrow$ 87.0% at $r_0=0.02$, while steady-state streaming falls to 30.2% ($T=32$) and to chance ($T=1$).

B. Centering, not dispersion, governs variability (E3)

Because dwell time is exponential in the barrier offset, “add dispersion” is ambiguous until one says which statistic is held fixed. Fig. 4 (left) shows both fabrication corners at fixed r_0 : with the *median* device fixed, accuracy degrades (72.0% $\rightarrow$ 67.5% at $r_0=0.05$, $\sigma_\Delta: 0 \rightarrow 1.5$); with the *mean* correlation time fixed, it rises dramatically (72.0% $\rightarrow$ 94.4%). The resolution is that mean- τ_{corr} centering is median centering plus a *uniform* rate multiplier $e^{\sigma_\Delta^2/2}$ ($3.1\times$ at $\sigma_\Delta = 1.5$) applied to every device. Substituting the median-device effective sampling ratio $r_0^{\text{eff}} = r_0 e^{\sigma_\Delta^2/2}$ into the dispersion-free baseline curve of Fig. 3 predicts the mean- τ_{corr} results to within a few points (e.g. $r_0^{\text{eff}} \approx 0.15$ predicts $\gtrsim 95\%$; measured 94.4%), so the uniform offset explains essentially all of the apparent rescue, with dispersion itself contributing a residual *penalty* of 1–5 points in both corners—the slow tail costs more than the fast tail buys. The engineering reading: a device specification for overclocked streaming must pin the *central fluctuation rate*, and because of the exponential barrier dependence, “mean τ_{corr} ” and “median device” differ by $e^{\sigma_\Delta^2/2}$ —a larger effect than the dispersion the specification would nominally

constrain [14]. Population-level diversity of superparamagnetic devices has been exploited as a computational resource [?]; here, for a fixed trained network, it is a second-order cost.

C. Correlation-aware training (E4)

Networks trained with the telegraph source active learn representations that survive staleness (Fig. 5). At $T=32$ streaming, the network trained at $r_0^{\text{train}}=0.2$ holds 84.6% at $r_0=0.02$ where the ideal-trained baseline retains 28.1%—a +56.5-point recovery costing 3.3 points at nominal speed (94.5% vs. 97.8%); the $r_0^{\text{train}}=0.5$ network trades less nominal accuracy (96.6%, a 1.2-point cost) for a smaller deep-overclock rescue. At $T=1$ the rescue is partial—the same $r_0^{\text{train}}=0.2$ network reaches 84.1% vs. the baseline’s 27.5% at $r_0=0.2$, but no trained network escapes the deep-overclock collapse at $T=1$ —so training and averaging compose rather than substitute. This extends multi-sample (“sample-aware”) training [19] along the temporal axis: what is learned here is robustness to *correlation between* samples, not benefit from their number.

D. Throughput and energy (E5)

Settle-mode throughput is bounded by $1/\rho = 0.25$ images per τ_{corr} regardless of read speed; streaming has no such bound but pays the collapse; correlation-aware training moves the streaming curve (Fig. 6). Model selection for the correlation-aware points uses the validation split only. Against the best rule-compliant operating point at each accuracy target, the frontier reaches: $2.0\times$ at 97% (streaming $T=2$, $r_0=2$: 97.3%), $4.0\times$ at 95% (correlation-aware single-shot streaming at $r_0=1$: 95.6%), and $8.0\times$ at 90% (correlation-aware single-shot at $r_0=0.5$: 92.5%). These are threshold statements on a finite grid—the conceded accuracy at the 90% target is 4.4 points below the bar-clearing rule point (96.9%)—and they assume readout electronics are not limiting. For a millisecond-class device ($\tau_{\text{corr}} \sim 10$ ms), the rule-compliant optimum is ~ 25 single-read classifications/s; the frontier makes ~ 100 /s available at $\geq 95\%$ accuracy and ~ 200 /s at $\geq 90\%$ from the

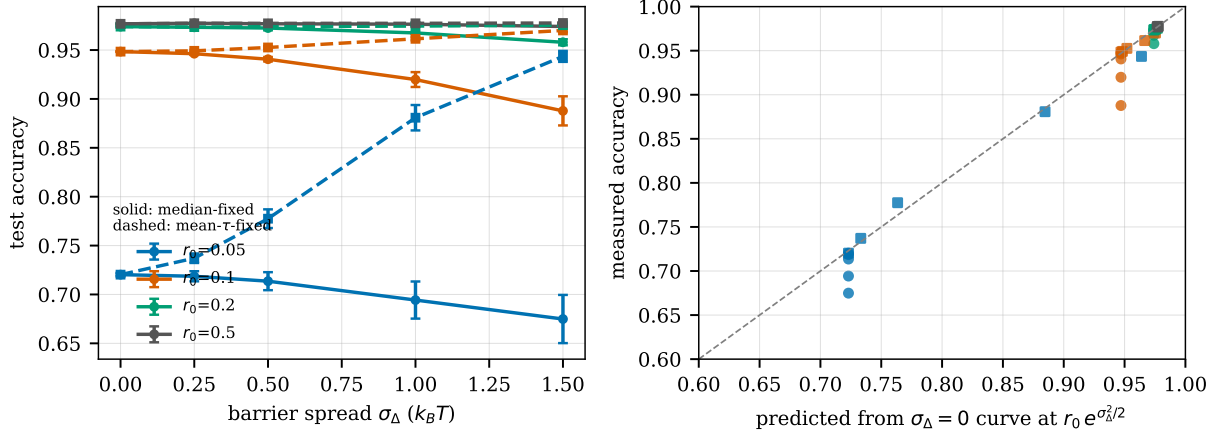


Fig. 4. **Left:** apparent effect of dwell-time dispersion depends on the fixed population statistic: median-fixed (solid) degrades; mean- τ_{corr} -fixed (dashed) appears to rescue ($T=32$, 5 chips, streaming). **Right:** the rescue is the centering’s uniform rate offset in disguise: accuracy is predicted by the baseline ($\sigma_\Delta = 0$) curve evaluated at the median-device effective ratio $r_0 e^{\sigma_\Delta^2/2}$ (mean- τ_{corr}) or r_0 (median), with dispersion contributing only a small residual penalty.

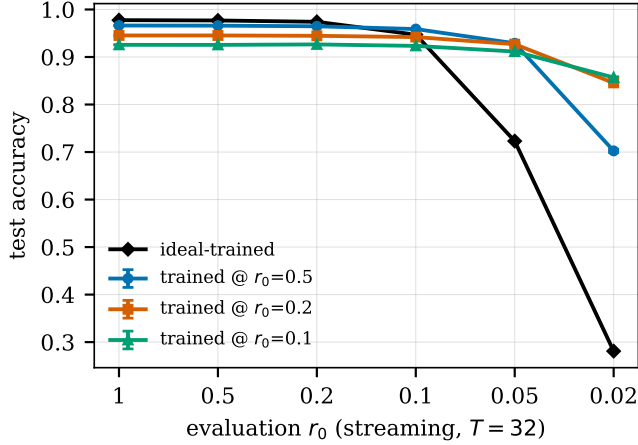


Fig. 5. Correlation-aware training flattens the streaming failure curve (steady-state, $T=32$; mean $\pm$ std over training and thermal seeds).

same silicon. Randomness-generation energy follows the bit count ($384T$ bits/inference): at $T=32$, 0.37 nJ from sMTJs vs. $37\text{--}369\text{ }\mu\text{J}$ from GPU/CPU PRNGs at the verified per-bit figures [1]—five to six orders of magnitude, unchanged by r_0 ; overclocking converts that energy advantage into throughput rather than surrendering it to the 2τ wait. An alternative decorrelation strategy—interleaving k devices per neuron round-robin—multiplies each device’s effective sampling interval by k , i.e. moves along the same accuracy-vs- r_0 curves at $k\times$ device area; our curves thus also quantify that area-throughput trade.

VI. DISCUSSION

Design guidance. (i) The $2\tau_d$ rule is the right specification for TRNG bitstream certification [1], [6], but for classification inference it is workload-dependent: batch-style (settle-mode) pipelines can read devices essentially arbitrarily fast, provided each input is given its $\approx 4\tau_{\text{corr}}$ settle—which becomes the

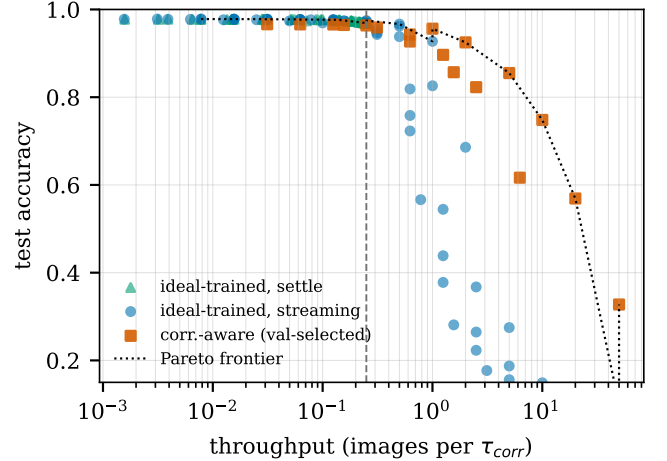


Fig. 6. Accuracy vs. physically-accounted throughput (settle mode pays its measured $\rho=4$ per input; dashed vertical: best rule-compliant single-read throughput $r_0=4$). Correlation-aware points use validation-selected training ratios.

real throughput cost. (ii) Streaming pipelines face a genuine boundary; it is moved by correlation-aware training and by sample averaging, not by fabrication uniformity. (iii) A device specification for overclocked operation must state *which* population statistic it pins: because dwell time is exponential in the barrier, mean- τ_{corr} and median-device conventions differ by $e^{\sigma_\Delta^2/2}$ in central rate—a larger effect than the dispersion the specification would nominally constrain. (iv) Barrier-type (Arrhenius) devices tolerate overclocking better than bias-independent-rate devices at matched τ_{corr} , because strong inputs buy fast refresh exactly where decisions are made.

Limitations. This is a simulation study with an idealized two-state model: no read-disturb, retention drift, or $1/f$ components [1], small MLPs, and MNIST-class benchmarks. Streaming results assume consecutive inputs are uncorrelated; correlated input streams (e.g. video) could interact with stal-

ness in either direction. The Pareto gains are grid-limited threshold statements, not continuous optima. Validation on fabricated arrays—e.g. the instrumented systems of [5]—and extension to convolutional networks and temperature drift are natural next steps.

VII. CONCLUSION

The price of violating the sMTJ sampling rule is not a wall but a fork. Batch-style inference with a measured $\approx 4\tau_{\text{corr}}$ per-input settle tolerates $200\times$ faster-than-rule sampling at under one accuracy point; steady-state streaming collapses to chance, is insensitive to fabrication-dispersion remedies (whose apparent effects reduce to the centering convention), and is largely rescued by correlation-aware training. Treating the sampling ratio as a design variable converts device auto-correlation into usable throughput—up to $2\text{--}8\times$ past the rule-compliant optimum depending on the accuracy target—while keeping the $\sim 10^6$ randomness-energy advantage that motivates p-bit hardware in the first place.

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